

# Does conditioning leave a structural memory?

## Large-area evidence for the dislocation picture

**Yinon Ashkenazy**

Racah Institute of Physics, The Hebrew University of Jerusalem

I. Popov (HUJI) · W. L. Millar, V. M. Bjelland, W. Wuensch (CERN)

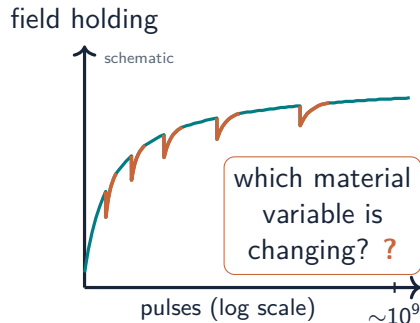
Montreux, Switzerland · Monday 5 October 2026



# Conditioning works. What does it change?



- ▶ Conditioning is how every high-gradient device reaches its usable field.<sup>a</sup>
- ▶ Breakdown rate has an extreme field dependence,  $\sim E^{30}$ .<sup>a</sup>
- ▶ It tracks the number of **high-field pulses**, not the number of breakdowns.<sup>a,b</sup>
- ▶ The field itself drives conditioning, not the arcs.<sup>a,b</sup>



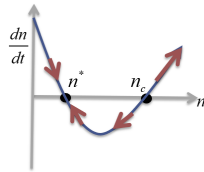
**Something in the metal stores the improvement. We do not know what.<sup>a</sup>**

Sources: <sup>a</sup>Wuensch et al., Rev. Mod. Phys. 98, 025004 (2026); <sup>b</sup>Degiovanni et al., PRAB 19, 032001 (2016).

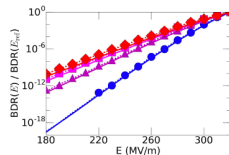
# Our model: breakdown as collective dislocation dynamics



- ▶ Field holding rises FCC→BCC→HCP<sup>d</sup> — the order of barriers to dislocation motion.
- ▶ The **MDDF** model: breakdown nucleates when the mobile-dislocation population, driven by Maxwell stress, runs away.<sup>a,b</sup>
- ▶ One model gets the  $\sim E^{30}$  rate and its temperature dependence,<sup>d</sup> and was shown to reproduce the histogram of dark-current spikes.<sup>c</sup>



mobile-dislocation runaway

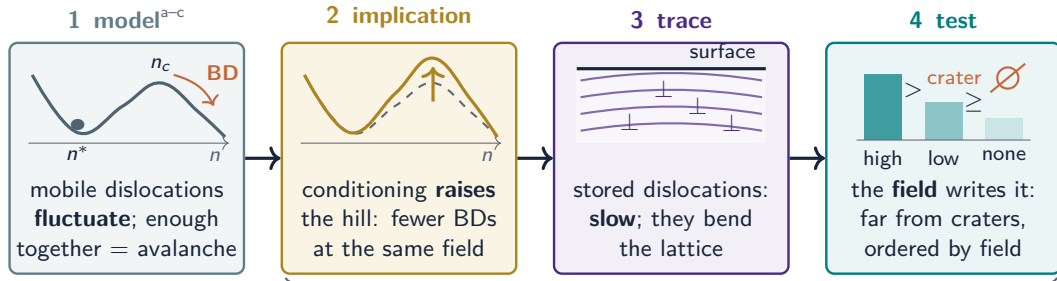


breakdown rate vs field

**Central prediction: conditioning *evolves the near-surface dislocation structure*.**<sup>d</sup>

Sources: <sup>a</sup>Engelberg et al., PRL 120, 124801 (2018); <sup>b</sup>PRAB 22, 083501 (2019); <sup>c</sup>PRAB 23, 123501 (2020); <sup>d</sup>Wuensch et al., RMP 98, 025004 (2026).

# What the model says the microscope should see



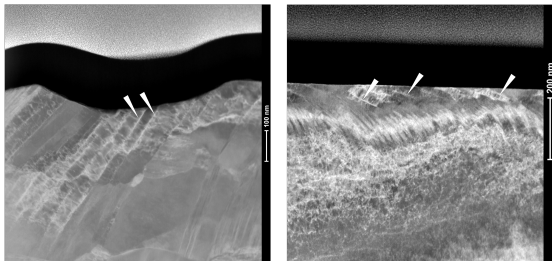
- ▶ In the model, a  $\sim 10^9$ -pulse memory must sit in stored, EBSD-visible dislocations.
- ▶ The field writes it: look far from craters; expect the order to follow the local field.

**predicts** where, and in which order

**does not predict** size, depth, or sign

Sources: <sup>a</sup>Engelberg et al., PRL 120, 124801 (2018); <sup>b</sup>PRAB 22, 083501 (2019); <sup>c</sup>PRAB 23, 123501 (2020); Wuensch et al., RMP 98, 025004 (2026).

# STEM already saw it, at specific locations



Cross-sectional STEM of conditioned hard (as-machined) Cu: dislocation-poor zone in the top  $\sim 200$  nm.<sup>a,b</sup>

- ▶ **Direct** evidence of near-surface plasticity in conditioned regions.<sup>a</sup>
- ▶ Fewer dislocation walls at 300 K and 30 K; denuded layer clearest at 30 K.<sup>a</sup>
- ▶ Fits the clue that soft Cu conditions far more slowly than hard Cu: the starting population matters.<sup>c</sup>
- ▶ FIB preparation plants its own dislocations; trust only large-scale structure.<sup>b</sup>
- ▶ One cross-section cannot speak for the whole electrode.

**Seen at specific locations. The large-area view was missing.**

Sources: <sup>a</sup>Jacewicz et al., JAP 137, 193302 (2025); <sup>b</sup>HUJI STEM (I. Popov), MeVArc 2025; <sup>c</sup>Korsbäck et al., PRAB 23, 033102 (2020).

# A sub-yield stress, repeated a billion times

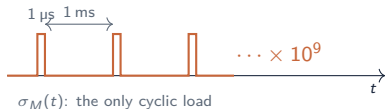


Maxwell stress

$$\sigma_M = \frac{\epsilon_0 E^2}{2}$$

at 80 MV/m: 0.028 MPa<sup>c</sup>

$\sim 10^3 \times$  below the yield of annealed Cu<sup>c</sup>



- ▶ Fatigue at  $10^8$ – $10^{10}$  cycles rearranges dislocations irreversibly, well below yield.<sup>a,b</sup>
- ▶ Here:  $\sim 10^9$  conditioning pulses.<sup>c</sup>
- ▶ Pulsed DC means **no RF pulsed heating** and no bulk currents. Maxwell stress is the only cyclic load.<sup>c</sup>
- ▶ MDDF needs only the stress that moves **mobile dislocations in a frustrated crystal**: far below yield.

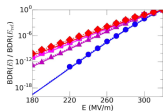
**The model predicts a large-area signature.**  
**Last year we asked for it; this year we looked.**

Sources: <sup>a</sup>Mughrabi, MMTB 40, 431 (2009); <sup>b</sup>Stanzl-Tschegg et al., Int. J. Fatigue 29, 2050 (2007); <sup>c</sup>Ashkenazy et al., arXiv:2606.19192 (2026).

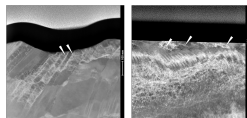
# Last year we asked for more forensics. Here it is.



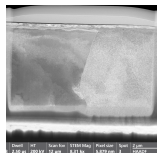
## MeVArc 2025



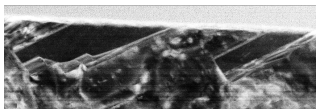
MDDF model



local STEM denuded zone<sup>a,b</sup>

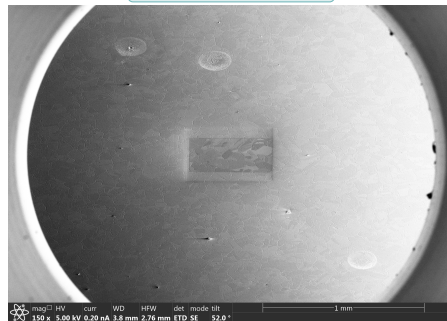


FIB lamella,  
HAADF overview<sup>a</sup>



cross-section STEM  
at the surface<sup>a</sup>

## MeVArc 2026

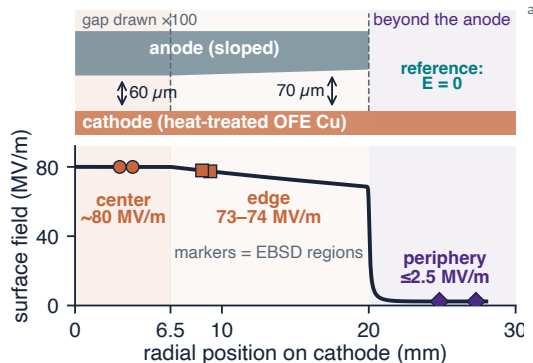


Large-area EBSD across one cathode

Closed with: other materials? more samples?

Sources: <sup>a</sup>HUJI STEM (I. Popov), MeVArc 2025; <sup>b</sup>Jacewicz et al., JAP 137, 193302 (2025).

# One cathode turns field exposure into a controlled coordinate



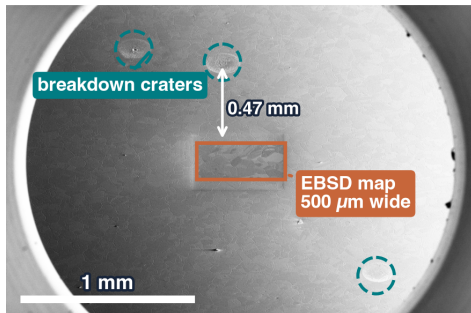
- ▶ Heat-treated OFE Cu, pulsed DC
- ▶ 1  $\mu\text{s}$  pulses at 1 kHz,  $\sim 10^9$  pulses
- ▶ Peak field  $\sim 80$  MV/m at center
- ▶ Nine EBSD regions: center, edge, periphery, and external reference

● high field (center, edge) ● periphery ( $\leq 2.5$  MV/m) ● unexposed reference

**The field gradient gives us internal controls on a single specimen.**

Sources: <sup>a</sup>geometry from Bjelland et al.; Ashkenazy et al., arXiv:2606.19192 (2026),  $E(r)$ : Fig. 3.

# EBSD maps plastic activity far from breakdown craters



- ▶ Large area where TEM could only sample a point
- ▶ 500  $\mu\text{m}$ -wide maps; 3  $\mu\text{m}$  step (lateral only)<sup>a</sup>; information depth  $\sim 40$  nm at 15 kV<sup>b,c</sup>
- ▶  $\sim 24,000$  indexed points per ROI<sup>a</sup>
- ▶ **LAM** = local average misorientation: lattice tilt between a point and its neighbors (KAM, LOS: cross-checks)<sup>a</sup>
- ▶ Misorientation is a proxy for GND density<sup>d</sup>

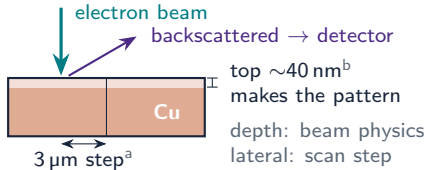
**We measure the cumulative response to the field, not crater damage.<sup>a</sup>**

Sources: <sup>a</sup>Ashkenazy et al., arXiv:2606.19192 (2026); <sup>b</sup> $\sim 40$  nm at 15 kV: CASINO Monte Carlo, Drouin et al., Scanning 29, 92 (2007); <sup>c</sup>measured 38–72 nm over 5–30 kV, Chen et al., Ultramicroscopy 111, 1488 (2011); <sup>d</sup>Pantleon, Scr. Mater. 58, 994 (2008).

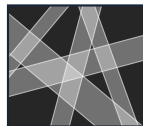
# What one EBSD pixel measures



## 1 15 kV electron beam, point by point



## 2 Pattern → lattice orientation



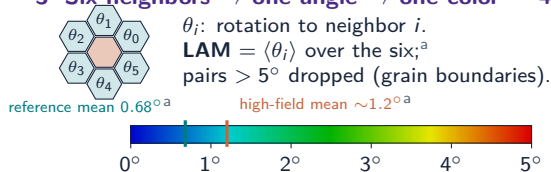
Kikuchi pattern



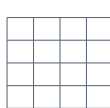
one lattice orientation for this point

$\sim 24,000 \text{ points}^a$  per region, unfiltered

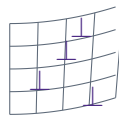
## 3 Six neighbors → one angle → one color



## 4 Why a rotation angle counts dislocations



flat:  $\theta = 0$



curved:  $\theta > 0$

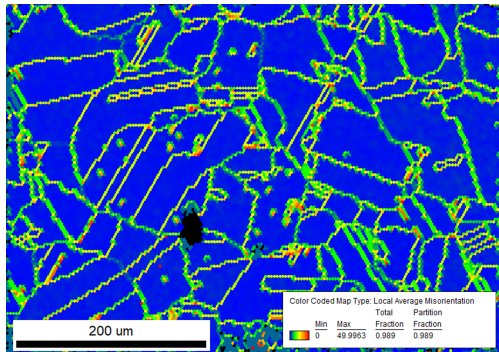
a dislocation proxy over  $500 \mu\text{m}$ , not a dislocation image

Bending a crystal by an angle over a distance needs **geometrically necessary dislocations**, so **LAM** is a proxy for GND density.<sup>c</sup>

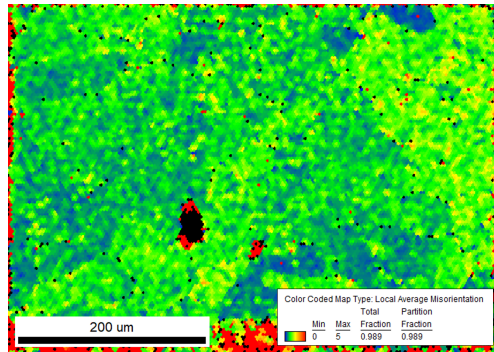
**Each  $3 \mu\text{m}$  point yields one angle, read from the top  $\sim 40 \text{ nm}$  of copper.<sup>b</sup>**

Sources: <sup>a</sup>Ashkenazy et al., arXiv:2606.19192 (2026); <sup>b</sup>information depth: Chen et al., Ultramicroscopy 111, 1488 (2011); <sup>c</sup>Pantleon, Scr. Mater. 58, 994 (2008).

# How to read a LAM map: wide-angle vs low-angle



0–50°: grain boundaries dominate

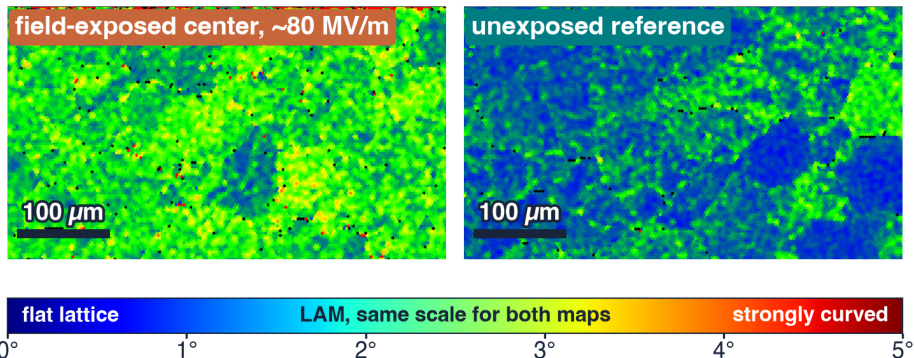


0–5° scale: intragrain curvature

**Rescaling to 0–5° exposes the intragrain misorientation we care about.**

Source: Ashkenazy et al., arXiv:2606.19192 (2026). Same area, two color scales.

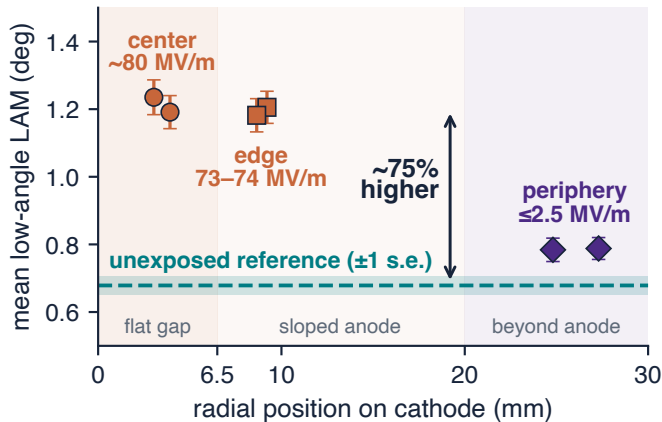
# High-field copper contains more intragrain curvature



**Same scale, same magnification: field-exposed grains carry more curvature.**

Source: Ashkenazy et al., arXiv:2606.19192 (2026). Both maps cropped to the same  $485 \times 279 \mu\text{m}$  field; LAM  $0-5^\circ$ .

# The signal separates into three exposure tiers

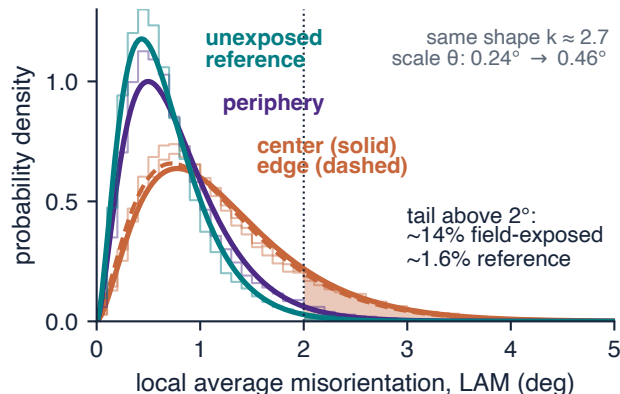


- ▶ Center and edge:  $\sim 1.2^\circ$
- ▶ External reference:  $\sim 0.68^\circ$
- ▶ High field is  $\sim 75\%$  higher
- ▶ Grain size does not follow this hierarchy

**The ordering follows field history, not sample position or grain size.**

Source: Ashkenazy et al., arXiv:2606.19192 (2026). Error bars: grain-corrected standard errors; band: unexposed reference  $\pm 1$  s.e.

# Field exposure builds a high-misorientation tail



**LAM tail above  $2^\circ$**

$\sim 0.14$  field-exposed  
 $\sim 0.016$  reference

$\sim 8\times$  **increase**

Gamma shape  $k \approx 2.7$  fixed; the scale roughly doubles.

**The whole distribution stretches; a uniform offset cannot do that.**

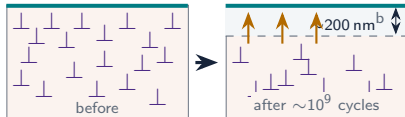
Source: Ashkenazy et al., arXiv:2606.19192 (2026). One ROI per tier (histograms faded, gamma fits solid, edge dashed); shaded: tail above  $2^\circ$ .

# The large-area test supports the model's prediction



## Hard, as-machined Cu

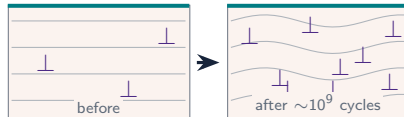
dense starting population (2025 STEM cathodes)



dislocations escape to the surface or annihilate  $\Rightarrow$  **depleted layer (STEM)**<sup>b</sup>

## Heat-treated OFE Cu

sparse starting population (this cathode)



one load  
 $\sigma_M = \frac{1}{2} \epsilon_0 E^2$   
 $\sim 10^9$  cycles  
 $10^{-3}$  of yield

the manuscript's interpretation<sup>a</sup>

same load generates and rearranges dislocations  $\Rightarrow$  **lattice curvature (EBSD, LAM  $\sim 75\%$  higher)**<sup>a</sup>

- To our knowledge, the first **large-area** observation that conditioned regions carry a different near-surface dislocation structure, ordered as MDDF predicts.<sup>a</sup>
- The measured order follows the simulated local field:  
center  $\approx$  edge  $>$  periphery  $>$  reference.<sup>a</sup>

**Before, dislocations were plausible.**

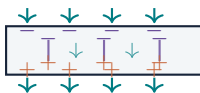
**Now, a dislocation-sensitive state maps the conditioning history.**

Sources: <sup>a</sup>Ashkenazy et al., arXiv:2606.19192 (2026), discussion; <sup>b</sup>Jacewicz et al., JAP 137, 193302 (2025).

# Elastic screening: dislocations respond like bound charge



## Dielectric in an applied field

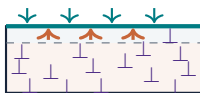


applied field  $E_0$   
bound charge polarizes  
interior field  
reduced to  $E_0/\epsilon_r$

**1**  $\lesssim 0.1 \text{ nm}$   
**Field penetration**  
Thomas–Fermi:  
the field stops at  
the surface.

**2**  $\ell_D \propto 1/\sqrt{\rho} \approx 25 \text{ nm}^a$  (order of magnitude)  
**Debye-like screening of fluctuations**  
At  $\rho \sim 10^{14} \text{ m}^{-2}$ , only dislocations within  
 $\sim \ell_D$  of the surface exchange fluctuating stress  
with it; deeper ones are driven by the load but  
silent at the surface. Observable: LAM; denuded  
 $\sim 200 \text{ nm} \approx 8 \ell_D$ .<sup>d</sup>

## Copper under the Maxwell traction



**1** field stops at the surface  
**3**  $\sigma_M$  enters as stress  
**2** dislocations as bound  
charge (dipole regime)<sup>b,c</sup>

**3**  $\sim \text{mm}$   
**Applied load: not screened**  
Dislocation stress is  
self-equilibrated; ordinary  
elasticity over the loading  
geometry.<sup>d</sup>

**A plasma of dipoles:** Debye-like for fluctuations, dielectric-like for the mean load; it reduces, it does not nullify.  
**The surface is an absorbing wall:** a sheath of  $\sim \ell_D$  sets what the surface sees, hence emission and breakdown.

**Weak, screened internal stresses set sources and structure inside the sheath; the strong external load drives mobile dislocations through it.**

Sources: <sup>a</sup>Groma et al., PRL 96, 165503 (2006); <sup>b</sup>Lemaître et al., PRE 104, 024904 (2021); <sup>c</sup>Livne et al., PRE 107, 055004 (2023); <sup>d</sup>Ashkenazy et al., arXiv:2606.19192 (2026).

# Copper establishes the effect, not universality



## What is established

- ▶ Large-area structural difference
- ▶ Three independent metrics
- ▶ Internal and external controls
- ▶ Supports the MDDF prediction

## What remains open

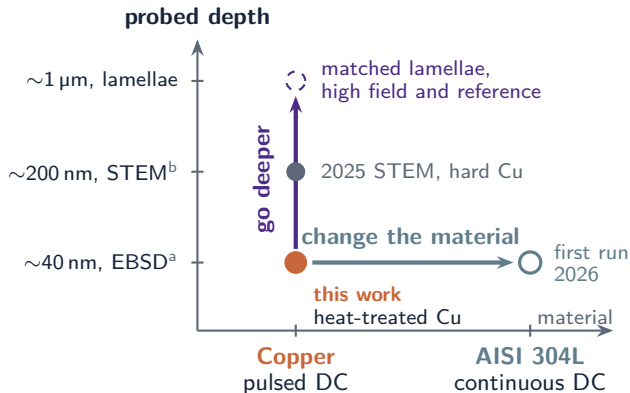
- ▶ One conditioned cathode, one experiment
- ▶ Full depth profile
- ▶ Grain-orientation dependence
- ▶ More conditioned cathodes
- ▶ Cross-material generality

**Limits of this dataset:** field level and breakdown density vary together by construction ( $\sim 24, 12, 5$  breakdowns per  $\text{cm}^2$ , center outward); LAM values are relative, compared only between regions mapped under the same step and protocol.

**A stronger test changes both the material and the loading history.**

Source: Ashkenazy et al., arXiv:2606.19192 (2026), discussion and limitations.

# Next: go deeper, and change the material



- ▶ Matched high-field and reference FIB lamellae; TEM, STEM and EBSD through the top  $\sim 1\ \mu\text{m}$
- ▶ Artifact controls from the 2025 “red herrings”
- ▶ One depth profile from STEM depletion to EBSD curvature?
- ▶ Stainless steel under continuous DC, with Consorzio RFX — next slide

**Depth continues the Uppsala STEM line; material tests the mechanism itself.**

Sources: <sup>a</sup>EBSD  $\sim 40\ \text{nm}$  at 15 kV: CASINO, Drouin 2007; <sup>b</sup>hard-Cu STEM zone  $\sim 200\ \text{nm}$ : HUJI, I. Popov 2025; interim work plan (2026).

# Steel: different material, different structural evolution



## What differs from copper

- ▶ Lower stacking-fault energy: planar slip, different dislocation mobility
- ▶ Continuous DC: a steady traction of  $\sim 16 \text{ kPa}^a$ , not  $\sim 10^9$  pulses
- ▶ Metastable austenite is a built-in strain gauge:  $\varepsilon$  and  $\alpha'$  martensite mark plastic strain above a few  $0.1\% ^b$

The dislocation picture expects a different near-surface evolution. The test: does any field-correlated change appear at all?



Consorzio RFX — De Lorenzi, Pilan, Spada

## First run with Consorzio RFX<sup>c</sup>

- ▶ AISI 304L at the RFX HVPTF: continuous DC, 90 h and 73 h at 59–61 MV/m; zero breakdowns, stopped at emission switch-on
- ▶ 26 EBSD maps (HUJI, I. Popov): the exposed apex lies inside or below the unexposed range
- ▶ Nothing was mapped before exposure, so this design cannot establish a field effect

**RFX HV program: N. Pilan, next talk (12:30).**

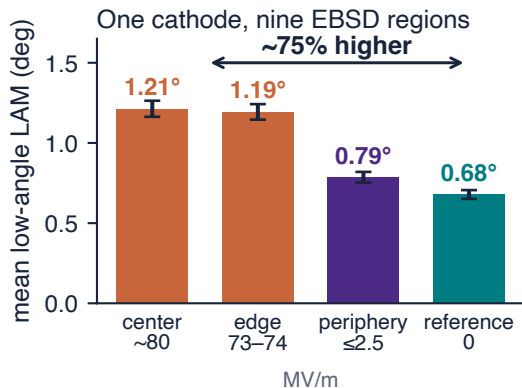
**Copper supports the memory; steel asks whether it is universal.**

Sources: <sup>a</sup> $\sim 16 \text{ kPa} = \varepsilon_0 E^2 / 2$  at 60 MV/m; <sup>b</sup>Bayerlein et al., MSEA 114, L11 (1989); <sup>c</sup>RFX project record, EBSD pass of 29 Sep 2026.

# Conditioning appears to leave a subsurface memory



1. Our model predicts conditioning evolves the near-surface dislocation structure.
2. 2025 STEM saw it locally; **large-area EBSD now supports it** ( $\sim 75\%$  higher misorientation).
3. Depth and stainless steel test whether the memory survives a change of scale, alloy, and loading.



**Last year: “other materials? more samples?” — here is the large-area answer.**

Copper result: Ashkenazy et al., arXiv:2606.19192 (2026).

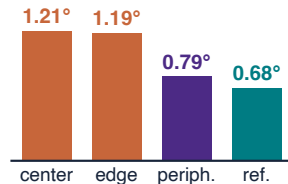
# Conclusions, and an invitation



1. The model predicted a structural memory of conditioning.
2. STEM saw it locally; large-area EBSD now supports it.
3. Open: depth<sup>a</sup>, replication, other materials — a stainless-steel test with Consorzio RFX is under way.<sup>b</sup>

**Talk to me: new materials, other electrode configurations, cross-facility comparisons, same-area before-and-after mapping — let us test this together.**

mean LAM, one cathode



[arxiv.org/pdf/2606.19192](https://arxiv.org/pdf/2606.19192)

Ashkenazy et al., arXiv:2606.19192

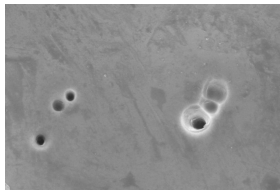
Sources: <sup>a</sup>Depth program: interim work plan (2026); <sup>b</sup>stainless-steel test: RFX project record (29 Sep 2026).

# Backup: RFX first look (preliminary)

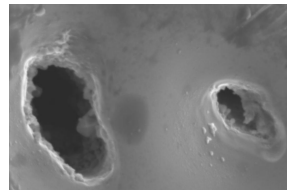


## Preliminary — no field effect claimed

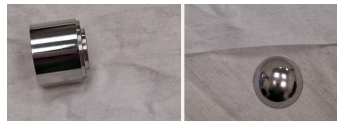
- ▶ AISI 304L at RFX HVPTF; continuous DC: 90 h (#3-25), 73 h (#4-25); peak 59–61 MV/m; zero breakdowns (stopped at emission switch-on)
- ▶ SEM: “exploded” near-surface inclusions (3–7  $\mu\text{m}$ ) on field-exposed samples; reference cavities round and clean<sup>a</sup>
- ▶ EBSD: #1-25, #2-25, #3-25 index (97–99.5%); #4-25: two maps, one an outlier on confidence index, position unrecorded
- ▶ Apex-vs-side contrast reproduces on never-installed #2-25: forming geography



#1-25 reference: round, clean (HFW 16.6  $\mu\text{m}$ )<sup>a</sup>



#3-25 field-exposed: “exploded” (HFW 10  $\mu\text{m}$ )<sup>a</sup>



Sectioned electrode

Selected panels; the full SEM set is in the RFX SEM report.

Sources: <sup>a</sup>SEM report, I. Popov (20 May 2026); RFX project record through 29 Sep 2026.

# Backup: RFX sample and data status (29 Sep 2026)



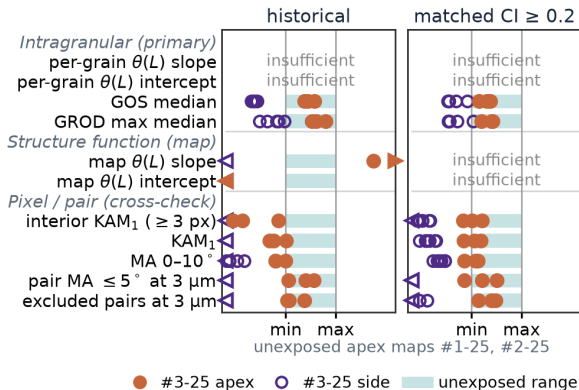
| Sample | Role          | Exposure                 | Status                                                        |
|--------|---------------|--------------------------|---------------------------------------------------------------|
| #1-25  | reference     | none                     | SEM; EBSD apex + side, 97% indexed                            |
| #2-25  | reference     | none                     | EBSD apex (18 Aug, 3 maps) + side; LAM done                   |
| #3-25  | field-exposed | 90 h DC, peak<br>61 MV/m | SEM; EBSD apex + side, 99.5% indexed;<br>LAM done             |
| #4-25  | field-exposed | 73 h DC, peak<br>59 MV/m | SEM/EDS; EBSD two maps, one an outlier<br>on confidence index |

- ▶ Received since July: conditioning report and  $E(\theta)$ ; alloy confirmed as 304L, mill certificate unrecoverable; electropolishing partly documented; all 26 catalog EBSD maps (36 scans), 28 Sep.
- ▶ XRD:  $\delta$ -ferrite content differs between the reference and exposed pairs, so the clean comparison is apex vs side within one electrode.
- ▶ Wide-vs-narrow map class = grain size and indexing quality (also on never-installed #2-25), not field.
- ▶ Still open: thermal history of #3-25 and #4-25 (vacuum firing?); matrix Cr/Ni EDS on all four under identical conditions; SEM  $\theta$ -survey as the first exposure marker.

# Backup: RFX apex vs unexposed range (29 Sep 2026)



Null so far — not a field effect, not proof of saturation

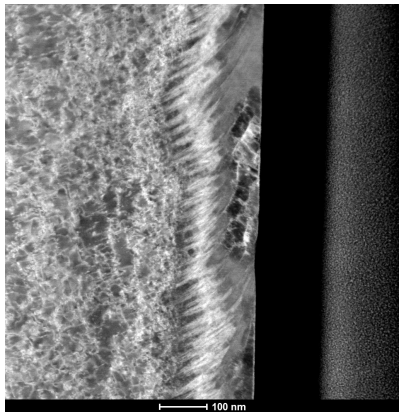


Wide map class, 2 of 4 panels (full board: RFX record). Only point above range: map  $\theta(L)$  slope ( $n = 2$  baseline).

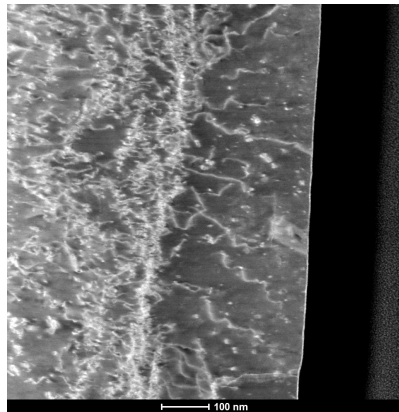
Source: RFX project record, orientation pass of 29 Sep 2026 (all 26 catalog maps); EBSD I. Popov, HUJI.

- ▶ Unit = one EBSD map; #3-25 apex vs same-class unexposed apex ( $n = 2-3$  vs  $3-4$ ).
- ▶ Inside or below the unexposed range (GOS, in-grain  $\theta(L)$  slope, interior KAM, histogram indicators), also at matched confidence index.
- ▶ Where it differs it is *cleaner*, not rougher ( $p \geq 0.10$ ).
- ▶ Power: 80% at 1 SD needs  $\approx 17$  maps per group; the data exclude only a uniform apex shift  $\gtrsim 2.5$  map-SD. Runs stopped at emission switch-on, zero breakdowns.

# Backup: field-exposed vs reference STEM (2025)



**Field-exposed, 300 K: fewer walls<sup>a,b</sup>**



**Reference, 300 K: walls to surface<sup>a,b</sup>**

Sources: <sup>a</sup>HUJI STEM (I. Popov), MeVArc 2025; <sup>b</sup>Jacewicz et al., J. Appl. Phys. 137, 193302 (2025), Figs. 8–10.